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REACTION ENGINEERING

CHE G641

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Non Ideal Reactors



Introduction

➤ Why this non-ideal flow important?

➤ So far, the reactors we have considered ideal flow patterns

- Residence time of all molecules are identical
- Perfectly mixed CSTRs & batch reactors
- No axial diffusion in a PFR/PBR
- But the reality is always in-between.

➤ Description of non ideal reactors

- Distribution of residence times
- Quality of mixing
- Models used to describe the system

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Ideal vs non ideal Reactors

Ideal Reactors

- Uniform flow distribution
- Uniform mixing
- Ideal mixing
- Ideal design equations
- No flow effects
- No bypass
- No radial variation in concentration
- No hot spots

Non-ideal Reactors

- Non-uniform flow
- In-efficient mixing, unmixed
- Design equations changes
- Different flow effects can occur:
Laminar & turbulent
- Bypass
- Affect the reactor design, the yield or temperature development of a chemical process

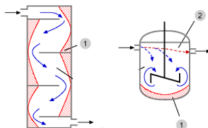
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Non ideal flow in a CSTR

- Ideal CSTR: uniform reactant concentration throughout the vessel
<https://www.youtube.com/watch?v=J6RgQ9gDDeU>
- Real stirred tank
 - Relatively high reactant concentration at the feed entrance
 - Relatively low concentration in the stagnant regions, called dead zones



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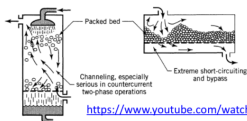
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Non ideal flow in PBRs

Ideal plug flow reactor: all reactant and product molecules at any given axial position move at same rate in the direction of the bulk fluid flow

Real plug flow reactor: fluid velocity profiles, turbulent mixing, & molecular diffusion cause molecules to move with changing speeds and in different directions



<https://www.youtube.com/watch?v=TCMUPNPWG1g>

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Mixing

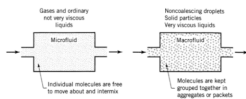
- Some fluid elements are coming and leaving-> spent different lengths of time
 - During this time, did they interact? Or simply leaving?
 - How these fluid elements are interacting in the reactor?
 - How much time each fluid element spent in the reactor?

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Micro & Macro mixing



Macromixing – distribution of residence times in the reactor

Micromixing – description of how molecules of different ages interaction with each other

- **Complete segregation** – all molecules of same age group remain together until they exit the reactor
- **Complete micromixing** – molecules of different age group are completely mixed

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Micromixing



The earliness or lateness of fluid mixing in the reactor

Whether the fluid is a micro or macro fluid



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Residence time distribution



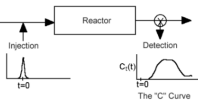
➤ Flow through a reactor is characterized by:

The amount of time molecules spend in the reactor, called the RTD

➤ $RTD = E(t)$ = "residence time distribution" function

RTD is measured experimentally by injecting an inert "tracer" at $t=0$ and measuring the tracer concentration $C(t)$ at the exit as a function of time

Tracer should be easy to detect & have physical properties similar to the reactant



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Residence time distributions

- Exit age distribution $E(t)dt$
 - The fraction of fluid element at the exit which has age between t and $t+dt$
- Cumulative residence time distribution, $F(t)$
 - Fraction of material in exit stream with age less than t
- Internal age distribution, $I(t)dt$
 - Fraction of material within vessel which has age between t and $t+dt$

$$F(t) = \int_0^t E(t)dt \quad \text{or} \quad \frac{dF(t)}{dt} = E(t), \quad 1 - F(t) = \frac{V}{F^1} I(t)$$

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Residence time distribution



- RTD $\equiv E(t) \equiv$ "residence time distribution" function
- RTD describes the amount of time molecules have spent in the reactor

$$E(t) = \frac{C(t)}{\int_0^\infty C(t)dt} = \frac{\text{tracer concentration at reactor exit between time } t \text{ and } t+\Delta t}{\text{sum of tracer concentration at exit for an infinite time}}$$

$$\int_0^\infty E(t)dt = 1 \quad \begin{aligned} E(t) &= 0 \text{ for } t < 0 \text{ since no fluid can exit before it enters} \\ E(t) &\geq 0 \text{ for } t > 0 \text{ since mass fractions are always positive} \end{aligned}$$

- Fraction of fluid element in the exit stream with age less than t_1 is:

$$\int_0^{t_1} E(t)dt$$

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Mean residence time

For an ideal reactor, the space time τ is defined as V/u_0
The mean residence time t_m is equal to τ in either ideal or nonideal reactors

$$t_m = \frac{\int_0^\infty t E(t) dt}{\int_0^\infty E(t) dt} = \frac{\int_0^\infty t E(t) dt}{1} = \tau = \frac{V}{u_0}$$

By calculating t_m , the reactor V can be determined from a tracer experiment

The spread of the distribution (variance):

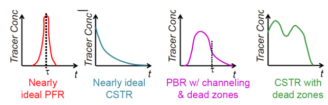
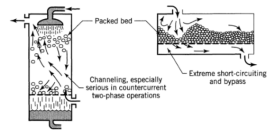
$$\sigma^2 = \int_0^\infty (t - t_m)^2 E(t) dt \quad \sigma^2 = \int_0^\infty t^2 E(t) dt - t_m^2$$

Skewness factor: whether the fluid is coming out much before or whether it is coming, much after the mean residence time

$$s^3 = \int_0^\infty (t - t_m)^3 E(t) dt / \sigma^{3/2}$$



RTD profiles





Cumulative RTD – F(t)

- $F(t)$ = fraction of effluent that has been in the reactor for less than

time

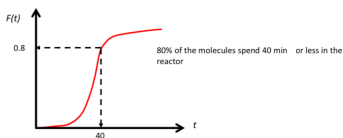
$$F(t) = \int_0^t E(t) dt$$

$$F(t) = 0 \text{ when } t < 0$$

$$F(t) \geq 0 \text{ when } t \geq 0$$

$$F(\infty) = 1$$

$$1 - F(t) = \int_t^{\infty} E(t) dt$$



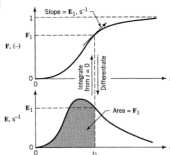
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Relationship between E & F curves

- $F(t)$ = fraction of effluent that has been in the reactor for less than

til



$$F(t) = \int_0^t E(t) dt$$

$$E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$$

- $E(t)$ = Fraction of material leaving reactor that was inside for a time between t_1 & t_2

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A pulse of tracer was injected into a reactor, and the effluent concentration as a function of time is in the graph below. Construct a figure of $C(t)$ & $E(t)$ and calculate the fraction of material that spent between 3 & 6 min in the reactor

t min	0	1	2	3	4	5	6	7	8	9	10	12	14
C g/m ³	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0

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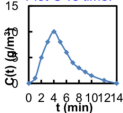
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t min	0	1	2	3	4	5	6	7	8	9	10	12	14
C g/m ³	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0

To tabulate $E(t)$: divide $C(t)$ by the total area under the $C(t)$ curve, which must be numerically evaluated as shown below:

Plot C vs time:



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$$\int_0^{\infty} C(t) dt = \int_0^{10} C(t) dt + \int_{10}^{14} C(t) dt$$

$$\int_{x_0}^{x_N} f(x) dx = \frac{\Delta t}{3} (f_0 + 4f_1 + 2f_2 + 4f_3 + 2f_4 \dots + 4f_{N-1} + f_N)$$

$$\int_{x_0}^{x_2} f(x) dx = \frac{\Delta t}{3} (f_0 + 4f_1 + f_2)$$

$$\rightarrow \int_0^{\infty} C(t) dt = 47.4 \frac{\text{g} \cdot \text{min}}{\text{m}^3} + 2.6 \frac{\text{g} \cdot \text{min}}{\text{m}^3} = 50 \frac{\text{g} \cdot \text{min}}{\text{m}^3}$$

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t min	0	1	2	3	4	5	6	7	8	9	10	12	14
C g/m ³	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0
E(t)	0	0.02	0.1	0.16	0.2	0.16	0.12	0.08	0.06	0.044	0.03	0.012	0

$$\int_0^{\infty} C(t) dt = 50 \frac{\text{g} \cdot \text{min}}{\text{m}^3}$$

Tabulate E(t): divide C(t) by the total area under the C(t) curve:

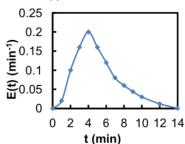
$$E(t) = \frac{C(t)}{\int_0^{\infty} C(t) dt}$$

$$E(t_0) = \frac{0}{50} = 0 \quad E(t_1) = \frac{1}{50} = 0.02$$

$$E(t_2) = \frac{5}{50} = 0.1 \quad E(t_3) = \frac{8}{50} = 0.16$$

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Plot E(t):



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RTD from the model

CSTR

$$E(t) = \frac{1}{\tau} \exp(-t/\tau)$$

PFR

$$E(t) = \delta(t - \tau), F(t) = H(t - \tau)$$

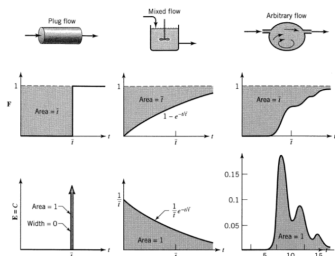
PFR- CSTR (or) CSTR- PFR

$$E(t) = \begin{cases} 0 & t < \tau_p \\ \frac{1}{\tau_r} \exp\left(-\frac{t - \tau_p}{\tau_r}\right) & t \geq \tau_p \end{cases}$$

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E & F curves for different reactors



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Reactor modeling with RTD

- When the fluid in a reactor is neither well mixed nor approximates plug flow (i.e., non-ideal), one can use RTD data and some model to predict conversion in the reactor

Zero adjustable parameters

- Segregation model & Maximum mixedness model

One parameter model

- Tanks-in-series model & dispersion model

Two adjustable parameters

- real reactor modeled as combinations of ideal reactors

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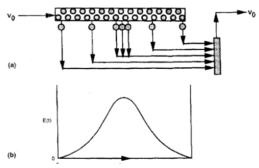


Segregation model

CSTR



PFR



- Flow is visualized in the form of globules and retain their identity
- Each globule consists of molecules of the same residence time
- Different globules have different residence times
- No interaction/mixing between different globules



Segregation model

$$\left[\begin{matrix} \text{Mean} \\ \text{conversion} \\ \text{of those globules} \\ \text{spending between} \\ \text{time } t \text{ and } t + dt \\ \text{in the reactor} \end{matrix} \right] = \left[\begin{matrix} \text{Conversion} \\ \text{achieved in a globule} \\ \text{after spending a time } t \\ \text{in the reactor} \end{matrix} \right] \times \left[\begin{matrix} \text{Fraction} \\ \text{of globules that} \\ \text{spend between } t \\ \text{and } t + dt \text{ in the} \\ \text{reactor} \end{matrix} \right]$$

The mean conversion is the average conversion of the various globules in the exit stream:

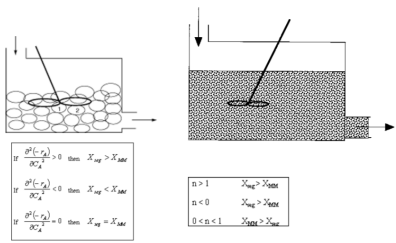
$$\bar{X}_A = \sum_j X_A(t_j) E(t_j) \Delta t$$

Conversion achieved after spending time t_j in the reactor

Fraction of globules that spend between t_j and $t_j + \Delta t$ in the reactor

$$\xrightarrow{\Delta t \rightarrow 0} \bar{X}_A = \int_0^{\infty} X_A(t) E(t) dt$$

$X_A(t)$ is from the **batch reactor** design equation

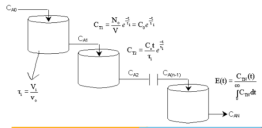


One parameter models

The non ideal tubular reactor is modeled as a series of identically sized CSTRs.

The RTD is analyzed in order to determine the number of ideal tanks in series and that gives approximately the same RTD as the non ideal reactor.

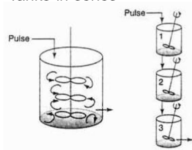
The RTD is analyzed from a tracer pulse injected into the first reactor of three equal sized CSTRs in series.



One parameter models



Tanks-in-series



$$C_1 = C_0 \exp(-vt/V_1) = C_0 \exp(-t/\tau_1)$$

$$C_2 = \frac{C_0 t \exp(-t/\tau_1)}{\tau_1}$$

$$C_3 = \frac{C_0 t^2 \exp(-t/\tau_1)}{2\tau_1^2}$$

For "N" tanks

$$E(t) = \frac{C_3(t)}{\int_0^\infty C_3(t) dt}$$

$$= \frac{t^2}{2\tau_1^3} e^{-t/\tau_1}$$

$$E(t) = \frac{t^{n-1}}{(n-1)! \tau_1^n} e^{-t/\tau_1}$$

$$E(\Theta) = \tau E(t) = \frac{n(n\Theta)^{n-1}}{(n-1)!} e^{-n\Theta}$$

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Tanks-in-series



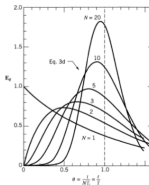
As the number of reactors becomes very large, the behavior of the system approaches the ideal plug flow reactor

Number of tanks in series can be calculate

$$\text{variance } \sigma_\theta^2 = \frac{\sigma^2}{\tau^2} = \int_0^\infty (\theta - 1)^2 E(\theta) d\theta$$

The number of tanks in series ($\tau = n \tau_i$)

$$n = \frac{1}{\sigma_\theta^2} = \frac{\tau^2}{\sigma^2}$$



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Dispersion model



Molecules will diffuse forward ahead of molar average velocity while others will lag behind.



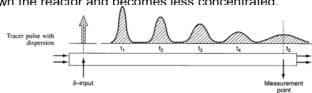
The one parameter to be determined in the dispersion model is the *dispersion coefficient*, D_a . The dispersion model is used most often for non-ideal tubular reactors. The dispersion coefficient can be found by a pulse tracer experiment.

Dispersion model



In this model, there is an axial dispersion of the material, resulting from molecular diffusion, i.e., $D_a \text{Ac} \left(\frac{\partial C}{\partial z} \right)$, superimposed on the bulk flow $v \text{Ac}$.

Dispersion causes the perfect tracer pulse to broaden as it moves down the reactor and becomes less concentrated.



Dispersion model



A mole balance on a particular component of the mixture over a short length ΔZ ,

$$-\frac{1}{A_z} \frac{dF_A}{dz} + r_A = 0$$

Here $F_A = vAC_A + AN_A$

$$\frac{D_z}{U} \frac{d^2 C_A}{dz^2} - \frac{dC_A}{dz} + \frac{r_A}{U} = 0$$

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Dispersion model



For a first order system, the analytical solution (Danckwerts, Chem. Eng. Sci., 2, 1 (1953)

$$X = 1 - \frac{4q \cdot \exp(Pe_r / 2)}{(1+q)^2 \exp(Pe_r q / 2) - (1-q)^2 \exp(-Pe_r q / 2)}$$

$$q = \sqrt{1 + 4Da / Pe_r}; Da = k \tau; \frac{\sigma^2}{\tau^2} = \frac{2}{Pe_r^2} (Pe_r - 1 + \exp(-Pe_r))$$

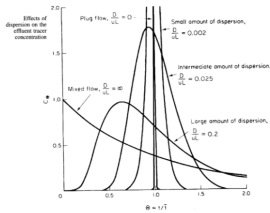
Pe_r is Pectlet No. = UL/D_a (1/ Pe_r = Bodenstein No)

Relationship between TIS & Dispersion model: $n \cong (Pe/2) + 1$

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Dispersion model



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Problem

A first order reaction $A \rightarrow B$ is carried out in a 10 cm dia tubular reactor with 6.36 m in length. $K = 0.25 \text{ min}^{-1}$. The results of the tracer test carried out on this reactor are provided

t(min)	0	1	2	3	4	5	6	7	8	9	10	12	14
C(mg/L)	0	1	5	8	10	8	6	4	3	2.2	1.5	0.6	0

- Calculate the conversion i) closed vessel dispersion model; ii) PFR;
iii) Single CSTR; iv) Tanks in series

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Problem

t(min)	C(t) (mg/L)	C(t)·delta t	E(t)	E(t)·t	E(t)·t·delta t	E(t)·t ²	E(t)·t ² ·delta t
0	0		0	0		0	
1	1	0.5	0.02	0.02	0.01	0.02	0.01
2	5	3	0.10	0.20	0.11	0.39	0.21
3	8	6.5	0.16	0.47	0.34	1.42	0.91
4	10	9	0.20	0.79	0.63	3.16	2.29
5	8	9	0.16	0.79	0.79	3.95	3.55
6	6	7	0.12	0.71	0.75	4.26	4.11
7	4	5	0.08	0.55	0.63	3.87	4.07
8	3	3.5	0.06	0.47	0.51	3.79	3.83
9	2.2	2.6	0.04	0.39	0.43	3.52	3.65
10	1.5	1.85	0.03	0.30	0.34	2.96	3.24
12	0.6	2.1	0.01	0.14	0.44	1.71	4.67
14	0	0.6	0.00	0.00	0.14	0.00	1.71
		50.65			5.13		32.24

$$\int_0^{\infty} C(t) dt \quad \tau = t_m = \int_0^{\infty} t \cdot C(t) dt \quad \int_0^{\infty} t^2 \cdot C(t) dt$$

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Solution

$$\begin{aligned} \tau = t_m &= \int_0^{\infty} t \cdot C(t) dt = \frac{V}{v} = 5.13 \text{ min}; \\ \int_0^{\infty} t^2 \cdot C(t) dt &= 32.24 \text{ min}^2; \\ \sigma^2 &= \int_0^{\infty} (t - \tau)^2 E(t) dt = \int_0^{\infty} t^2 E(t) dt - \tau^2 = 32.24 - (5.13)^2 = 5.92 \text{ min}^2; \\ \frac{\sigma^2}{\tau^2} &= \frac{5.92}{5.13^2} = 0.225 = \frac{2}{Pe_r^2} (Pe_r - 1 + \exp(-Pe_r)) \Rightarrow Pe_r \approx 7.7; \end{aligned}$$

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Solution

$$Da = \tau k = (5.13 \text{ min}) \cdot (0.25 \text{ min}^{-1}) = 1.28;$$

$$q = \sqrt{1 + 4Da / Pe_r} = \sqrt{1 + 4 \cdot \frac{(1.28)}{(7.7)}} = 1.29;$$

$$\frac{Pe_r \cdot q}{2} = \frac{(7.7) \cdot (1.29)}{2} = 4.97;$$

$$X = 1 - \frac{4 \cdot (1.29) \cdot \exp(7.7/2)}{(1 + 1.29)^2 \exp(4.97) - (1 - 1.29)^2 \exp(-4.97)} = 0.68;$$

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Solution

If the system is ideal PFR

$$X = 1 - \exp(-\tau k) = 1 - \exp(-Da) = 1 - \exp(-1.28) = 0.72$$

Ideal CSTR

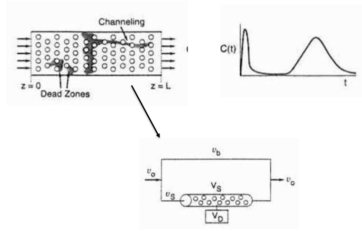
$$X = \frac{\tau k}{1 + \tau k} = \frac{Da}{1 + Da} = \frac{1.28}{1 + 1.28} = 0.56$$

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Two parameter models



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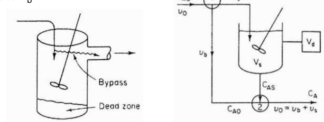
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Two parameter models

A real CSTR is modeled as a combination of an ideal CSTR of volume V_s , a dead zone V_d , and a bypass with a volumetric flow rate v_b



$$C_A = \frac{v_b C_{A0} + C_{As} v_s}{v_b + v_s} = \frac{v_b C_{A0} + C_{As} v_s}{v_0}$$
$$[C_{A0} v_b + C_{As} v_s] = [C_A (v_b + v_s)]$$
$$C_A = \beta C_{A0} + (1 - \beta) C_{As} \quad \alpha = V_s/V \text{ and } \beta = v_b/v_0$$

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Balance for the reactor

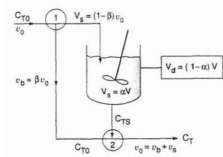
$$v_s C_{A0} - v_s C_{As} - k C_{As} V_s = 0$$

$$C_{As} = \frac{C_{A0}(1-\beta)v_0}{(1-\beta)v_0 + \alpha V k}$$

$$\frac{C_A}{C_{A0}} = 1 - X = \beta + \frac{(1-\beta)^2}{(1-\beta) + \alpha \tau k}$$

α & β parameters can be found from tracer tests

Two parameter models- Modeling with ideal reactors- Tracer test



$$\frac{C_{T1}}{C_{T0}} = 1 - \exp \left[-\frac{1-\beta}{\alpha} \left(\frac{t}{\tau} \right) \right]$$

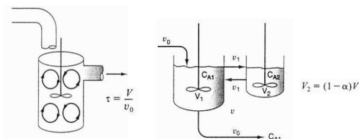
$$\frac{C_T}{C_{T0}} = 1 - (1-\beta) \exp \left[-\frac{1-\beta}{\alpha} \left(\frac{t}{\tau} \right) \right]$$

$$\ln \frac{C_{T0}}{C_{T0} - C_T} = \ln \frac{1}{1-\beta} + \left(\frac{1-\beta}{\alpha} \right) \frac{t}{\tau}$$



Real CSTR- two ideal CSTRs with interchange

In a real CSTR- highly agitated region in the vicinity of the impeller, outside this, there is a less agitation



$$C_{A1} = \frac{C_{A0}}{1 + \beta + \alpha \tau k - \{\beta^2 / [\beta + (1 - \alpha) \tau k]\}}$$

$$X = 1 - \frac{C_{A1}}{C_{A0}} = \frac{(\beta + \alpha \tau k)[\beta + (1 - \alpha) \tau k] - \beta^2}{(1 + \beta + \alpha \tau k)[\beta + (1 - \alpha) \tau k] - \beta^2}$$

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Real CSTR- two ideal CSTRs with interchange – Tracer Expts

$$\text{Reactor- 1: } V_1 \frac{dC_{T1}}{dt} = v_1 C_{T2} - (v_0 C_{T1} + v_1 C_{T1})$$

$$\text{Reactor-2: } V_2 \frac{dC_{T2}}{dt} = v_1 C_{T1} - v_1 C_{T2}$$

$$\tau \alpha \frac{dC_{T1}}{dt} = \beta C_{T2} - (1 + \beta) C_{T1}$$

$$\tau (1 - \alpha) \frac{dC_{T2}}{dt} = \beta C_{T1} - \beta C_{T2}$$

$$\left(\frac{C_{T1}}{C_{T10}} \right)_{pulse} = \frac{(\alpha m_1 + \beta + 1) e^{m_2 t / \tau} - (\alpha m_2 + \beta + 1) e^{m_1 t / \tau}}{\alpha (m_1 - m_2)}$$

where

$$m_1, m_2 = \left[\frac{1 - \alpha + \beta}{2\alpha(1 - \alpha)} \right] \left[-1 \pm \sqrt{1 - \frac{4\alpha\beta(1 - \alpha)}{(1 - \alpha + \beta^2)}} \right]$$